

## PROJECT AND TEAM INFORMATION

### Project Title

(Try to choose a catchy title. Max 20 words).

***Cipher Valley: A Gamified Environment for Learning Programming and Developing Computational Problem-Solving Skills***

### Student / Team Information

|                                                                                                                                        |                                                                                     |
|----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <b>Team Name:</b><br><b>Team #:</b>                                                                                                    | <b>Cipher Valley</b><br><b>TEAM NO-T334</b>                                         |
| <b>Team member 1 (Team Lead)</b><br><b>NAME-</b> ABHISHAL SINGH<br><b>STUDENT ID-</b> 2510360145<br><b>EMAIL-</b> 2510360145@geu.ac.in |   |
| <b>Team member 2</b><br><b>NAME-</b> NAVNEET RANA<br><b>STUDENT ID-</b> 2510012635<br><b>EMAIL-</b> 2510012635@geu.c.in                |  |
| <b>Team member 3</b><br><b>NAME-</b> BHAVSHAYA BHARDWAJ<br><b>STUDENT ID-</b> 2510012454<br><b>EMAIL-</b> [Enter College Email]        | [Enter Last Name, First Name]<br>[Enter Student ID]<br>[Enter College Email]        |
| <b>Team member 4</b><br>(Last Name, name: student ID: email, picture):                                                                 | [Enter Last Name, First Name]<br>[Enter Student ID]<br>[Enter College Email]        |

## PROPOSAL DESCRIPTION (10 pts)

### Motivation (1 pt)

*(Describe the problem you want to solve and why it is important. Max 300 words).*

*Today, programmers have the capabilities and the analytical skills necessary for a good learning experience, but the classic lecture method might make the learning a little monotonous and tiresome; therefore, writers may find it hard to keep the motivation and interest while practising on concepts of programming.*

*Cipher Valley works to combat this problem by using a game-based style of learning programming. This involves using programming puzzles along with quests, storytelling, levelling up, hints in context, and other feedback to make the educational process more interesting.*

*From the success of story-driven games like Grand Theft Auto V, Minecraft, and Stardew Valley, we asked ourselves, what if we put this concept into a programming classroom?*

*The project thus explores whether integrating Programmer's Workbench with this interactive progression, contextual hints and feedback, and adjusting problem difficulty accordingly, creates a more engaging experience for programmers new to programming.*

### State of the Art / Current solution (1 pt)

*(Describe how the problem is solved today (if it is). Max 200 words).*

*Minecraft Education Edition aggregates problem-solving, programming, and other educational experiences within a digital game platform. Baba Is You is a puzzle game where logic and reasoning are used to navigate rule-based scenarios. Prodigy Math gamifies mathematical practice through quests, recognitions, and levels.*

*Interest has also increased in the use of gamification techniques to make programming and other learning activities more fun. Many current platforms and games exemplify the use of design for playing and game mechanics to facilitate learning.*

*Other techniques are illustrated (besides gamification) like: interactive classes, practice, application in real life, quiz, group discussion and competing games.*

*Cipher Valley has plans to integrate some of these into a single game-based setting, where obviously the focus will be more on making programming more of a practical, contextual and fun experience.*

## Project Goals and Milestones (2 pts)

(Describe the project general goals. Include initial milestones as well any other milestones. Max 300 words).

The primary goal of Cipher Valley is to design and develop a functional prototype of a gamified programming-learning environment and investigate its potential to improve learner engagement and problem-solving experience.

- **Milestone 1 — Research and Requirements:** Review relevant literature, identify the problem, define research questions, and establish educational and technical requirements.
- **Milestone 2 — System and Game Design:** Design the game mechanics and learning system to effectively combine game-based interaction with programming education.
- **Milestone 3 — Designing the story base:** Develop an interactive storyline that supports game progression while maintaining learner engagement.
- **Milestone 4 — Learning System & C++/DSA Design:** Decide which programming concepts, challenges, C++ components, graphs, BFS/DFS/Dijkstra, etc. fit into the game and where they will be used.
- **Milestone 5 — Prototype development:** Develop a functional prototype based on the research, requirements, and system design established in the previous milestones.
- **Milestone 6 — Testing and Evaluation:** Test the prototype with users and collect measures such as challenges attempted and completed, attempts, hint usage, completion time, and structured feedback.
- **Milestone 7 — Analysis and Documentation:** Analyse results, document limitations and findings, and prepare the final report, presentation, demonstration, and GitHub documentation.

## Project Approach (3 pts)

(Describe how you plan to articulate and design a solution. Including platforms and technologies that you will use. Max 300 words).

The project will follow an iterative research-and-development approach. The first stage will involve reviewing literature on gamification, game-based learning, and programming education to establish the research foundation. Requirements will then be defined for both educational and technical components.

The prototype will be designed as a browser-based 2D interactive environment. Phaser.js and JavaScript will be used for the game layer, with HTML/CSS supporting the interface. Initial learning content will focus on variables, conditional statements, loops, functions, and basic debugging.

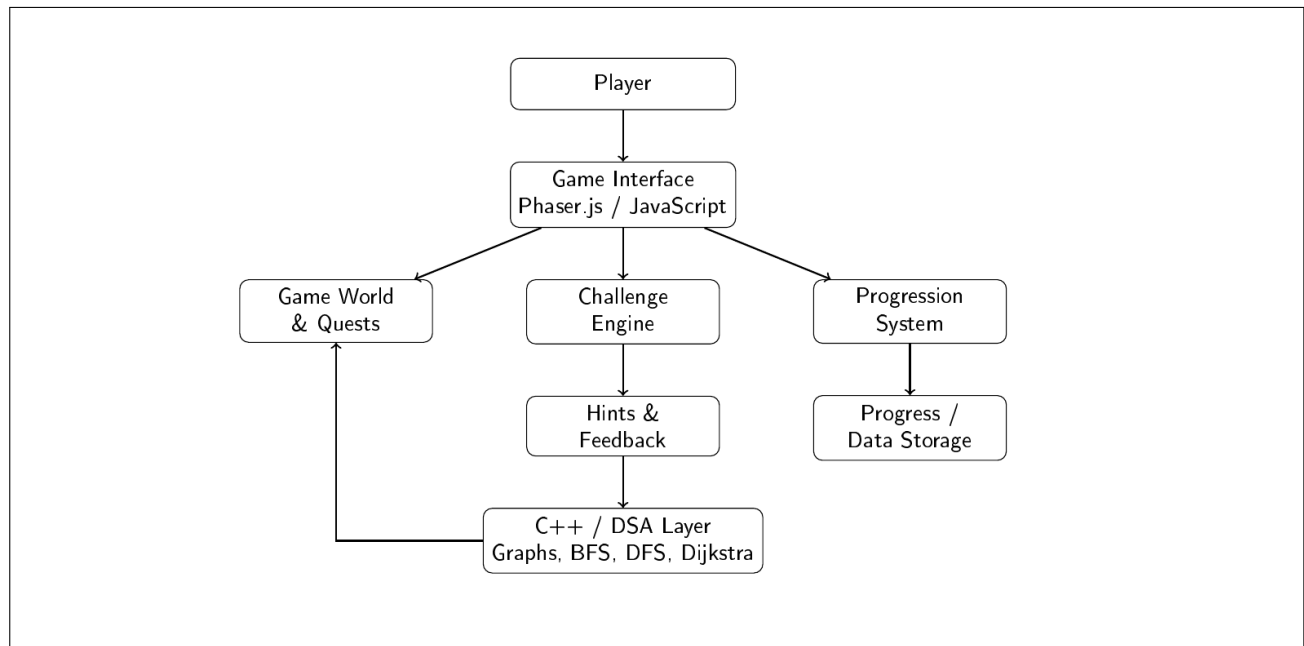
Programming challenges will be integrated into quests and game progression. A challenge engine will present problems and process submissions, while a hint and feedback mechanism will provide guidance without immediately revealing solutions. A progression system will record completed challenges and unlock subsequent content.

C++ will be incorporated for algorithmic and Data Structures and Algorithms components. Graph-based representations can be used for world navigation with BFS, DFS, and Dijkstra's algorithm applied where appropriate. JSON or SQLite may be used for structured game and progress data. Git and GitHub will be used for version control and documentation.

After implementation, user testing will collect quantitative interaction data and structured feedback. Results will be analysed to evaluate engagement and problem-solving experience and to identify potential benefits and limitations.

## System Architecture (High Level Diagram)(2 pts)

(Provide an overview of the system, identifying its main components and interfaces in the form of a diagram using a tool of your choice).



## Project Outcome / Deliverables (1 pts)

(Describe what are the outcomes / deliverables of the project. Max 200 words).

- The project should provide a working prototype of the Cipher Valley concept that exhibits the pedagogical advantages of programming in a game-based environment. It should feature a 2D interactive environment, programming challenges, quests or progression, contextual hints, instant feedback and simple means of monitoring progress.
- There will be opportunity to learn how to use C++, Data Structures and Algorithms, and game-development concepts in an interactive learning environment.
- Important outputs will be a maintained GitHub repository, documentation, the working prototype and a final report including the development process, results and limitations/discussions and lessons learnt.

## Assumptions

*( Describe the assumptions ( if any ) you are making to solve the problem. Max 100 words )*

*The project assumes that initial target users have basic computer literacy and are beginners or early-stage programming learners. Users are assumed to have access to a computer with a modern web browser. The target user has prior interaction with games or enjoys playing them. The results of the prototype will be considered as a benchmark rather than a real-world application.*

## References

*(Provide a list of resources or references you utilised for the completion of this deliverable. You may provide links).*

- 1. University referenced books for DSA integration and C++ programming, suggested by the university for learning the basic and fundamental as well as advanced topics of the subject.*
- 2. Gamified learning in higher education by Subash and Cundey (2018): <https://www.sciencedirect.com/science/article/abs/pii/S0747563218302541>*
- 3. The effectiveness of game introduction in programming by Guo and Lan: [https://www.researchgate.net/publication/379980212\\_Investigating\\_the\\_influence\\_of\\_gamification\\_on\\_motivation\\_and\\_learning\\_outcomes\\_in\\_online\\_language\\_learning](https://www.researchgate.net/publication/379980212_Investigating_the_influence_of_gamification_on_motivation_and_learning_outcomes_in_online_language_learning)*
- 4. Impact of Video Games on Problem-Solving and Logical Reasoning Abilities among Youth: [https://www.researchgate.net/publication/391256949\\_Impact\\_of\\_Video\\_Games\\_on\\_Problem-Solving\\_and\\_Logical\\_Reasoning\\_Abilities\\_among\\_Youth](https://www.researchgate.net/publication/391256949_Impact_of_Video_Games_on_Problem-Solving_and_Logical_Reasoning_Abilities_among_Youth)*